Physics 468/668: Statistical Physics **Spring 2019**

Lecture 21: Ising model in the first approximation

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21.1 Bethe approximation

In the Bethe approximation, a given spin σ_0 is regarded a the central member of a group, which consists of this spin and it q nearest neighbors. The Hamiltonian of this system under a mean molecular field B' is

$$H_{q+1} = -\mu B\sigma_0 - \mu(B+B')\sum_{j=1}^{q}\sigma_j - J\sum_{j=1}^{q}\sigma_0\sigma_j \tag{21.1}$$

where B the external magnetic field acting on the lattice. The internal field B' is determined by the condition of self-consistency, **which requires that the mean value, $\bar{\sigma_0}$, of the central spin be the same as the mean value, $\bar{\sigma_j}$, of any of the q neighbors**.

The partition function Q of this group of spins as a whole is given by

$$\begin{aligned} Q &= \sum_{\sigma_0,\sigma_j=\pm1}\exp\left[\frac{1}{kT}\left\{\mu B\sigma_0 + \mu(B+B')\sum_{j=1}^{q}\sigma_j + J\sum_{j=1}^{q}\sigma_0\sigma_j\right\}\right] \\ &= \sum_{\sigma_0,\sigma_j=\pm1}\exp\left[\alpha\sigma_0 + (\alpha+\alpha')\sum_{j=1}^{q}\sigma_j + \gamma\sum_{j=1}^{q}\sigma_0\sigma_j\right] \end{aligned} \tag{21.2}$$

where

$$\alpha = \frac{\mu B}{kT} \quad \alpha' = \frac{\mu B'}{kT} \quad \gamma = \frac{J}{kT} \tag{21.3}$$

Now, the right side of Eq 21.1 can be written as a sum of two terms, one pertaining to $\sigma_0 = \pm 1$ and the other to $\sigma_0 = -1$, that is

$$Q = Q_+ + Q_- \tag{21.4}$$

where

$$\begin{aligned} Q_\pm &= \sum_{\sigma_j=\pm1}\exp\left[\pm\alpha + (\alpha+\alpha'\pm\gamma)\sum_{j=1}^{q}\sigma_j\right] \\ &= e^{\pm\alpha}[2\cosh(\alpha+\alpha'\pm\gamma)]^q \end{aligned} \tag{21.5}$$

The mean value of the central spin is

$$\bar{\sigma_0} = \frac{Q_+ - Q_-}{Q} \tag{21.6}$$

While the mean value of any one of its q neighbors is given by

$$\begin{aligned}\bar{\sigma}_j &= \frac{1}{q}\overline{\left(\sum_{j=1}^{q}\sigma_j\right)} = \frac{1}{q}\left(\frac{1}{Q}\frac{\partial Q}{\partial \alpha'}\right) \\ &= \frac{1}{Q}[Q_+ \tanh(\alpha+\alpha'+\gamma) + Q_- \tanh(\alpha+\alpha'-\gamma)]\end{aligned} \tag{21.7}$$

$$Q_+[1-\tanh(\alpha+\alpha'+\gamma)] = Q_-[1+\tanh(\alpha+\alpha'-\gamma)] \tag{21.8}$$

Therefore, we finally obtain

$$e^{2\alpha'} = \left[\frac{\cosh(\alpha+\alpha'+\gamma)}{\cosh(\alpha+\alpha'-\gamma)}\right]^{q-1} \tag{21.9}$$

To study the possibility of spontaneous magnetization, we get $\alpha = \mu B/kT = 0$. Equation then reduces to

$$\alpha' = \frac{q-1}{2}\ln\left\{\frac{\cosh(\alpha'+\gamma)}{\cosh(\alpha'-\gamma)}\right\} \tag{21.10}$$

If $\gamma = 0$, α' is zero. Otherwise, α' may be zero unless γ exceeds a certain critical value (γ_c). To determine this value, we do Taylor expansion around $\alpha' = 0$, which gives

$$\alpha' = (q-1)\tanh\gamma\left\{\alpha' - \operatorname{sech}^2\gamma\frac{\alpha'^3}{3}\right\} \tag{21.11}$$

For all γ, $\alpha' = 0$ is one solution (similar to what we did in the previous lecture). A non zero solution requires that,

$$(q-1)\tanh\gamma > 1 \tag{21.12}$$

That is

$$\gamma > \gamma_c = \tanh^{-1}\left(\frac{1}{q-1}\right) = \frac{1}{2}\ln\left(\frac{q}{q-2}\right) \tag{21.13}$$

This corresponding temperature is,

$$T < T_c = \frac{2J}{k}/\ln\left(\frac{q}{q-2}\right) \tag{21.14}$$

which determines the Curie temperature of the lattice. We also infer that when T is close to T_c from the left side,

$$\begin{aligned}\alpha' &\approx (3\cosh^2\gamma_c[(q-1)\tanh\gamma - 1])^{1/2} \\ &\approx [3(q-1)(\gamma-\gamma_c)]^{1/2} \\ &\approx [3(q-1)\frac{J}{kT_c}(1-\frac{T}{T_c})]^{1/2}\end{aligned} \tag{21.15}$$

The parameter $\bar{L}$ which is a measure of the long-range order in the system.

$$\bar{L} = \frac{(Q_+/Q_-)-1}{(Q_+/Q_-)+1} = \frac{\sinh(2\alpha+2\alpha')}{\cosh(2\alpha+2\alpha')+\exp(-2\gamma)} \tag{21.16}$$

In the limit $B \to 0$, when T is approaching to T_c from the left side,

$$\bar{L_0} = \frac{\sinh(2\alpha')}{\cosh(2\alpha') + \exp(-2\gamma)} \approx \frac{2\alpha'}{1 + (q-2)/q} = \frac{q}{q-1}\alpha' \tag{21.17}$$

$$\bar{L_0} \approx \left[3\frac{q}{q-1} \left\{ \frac{q}{2} \ln\left(\frac{q}{q-2} \right) \right\} \left(1 - \frac{T}{T_c} \right) \right]^{1/2} \tag{21.18}$$

We see that, when $q \gg 1$, the equation reduce to their zeroth-order approximation $[3(1 - \frac{T}{T_c})]^{1/2}$.

21.2 The correlations between spin configurations

We shall now study correlations that might exist among neighboring spins in the lattice. For this, we evaluate the numbers $\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-}$ in terms of the parameters α, α' and γ, and compare the resulting expressions with the ones obtained under the mean field approximation. Carrying out summations in (2) over all the spins (of the group) except α_0 and σ_1, we obtain

$$Q = \sum_{\sigma_0, \sigma_1 = \pm 1} \left[\exp\{\alpha\sigma_0 + (\alpha + \alpha')\sigma_1 + \gamma\sigma_0\sigma_1\} \left\{ 2\cosh(\alpha + \alpha' + \gamma\sigma_0) \right\}^{q-1} \right] \tag{21.19}$$

Writing this as a sum of three parts pertaining, respectively, to the case (i) $\sigma_0 = \sigma_1 = \pm 1$, (ii) $\sigma_0 = -\sigma_1 = -1$, and (iii) $\sigma_0 = -\sigma_1 = \pm 1$, we have

$$Q = Q_{++} + Q_{--} + Q_{+-} \tag{21.20}$$

where,

$$\begin{aligned} \bar{N}_{++} &\propto e^{(2\alpha + \alpha' + \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \\ \bar{N}_{--} &\propto e^{(-2\alpha - \alpha' + \gamma)} [2\cosh(\alpha + \alpha' - \gamma)]^{q-1} \\ &= e^{(-2\alpha - 3\alpha' + \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \end{aligned} \tag{21.21}$$

and

$$\begin{aligned} \bar{N}_{+-} &\propto e^{(-\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} + e^{(\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \\ &= 2e^{(-\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \end{aligned} \tag{21.22}$$

Normalizing these expressions with the help of the relationship

$$\bar{N}_{++} + \bar{N}_{--} + \bar{N}_{+-} = \frac{1}{2} qN \tag{21.23}$$

we obtain the desired results

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-}) = \frac{1}{2} qN \frac{(e^{2\alpha + 2\alpha' + \gamma}, e^{-2\alpha - 2\alpha' + \gamma}, 2e^{-\gamma})}{2\{e^{\gamma}\cosh(2\alpha + 2\alpha') + e^{-\gamma}\}} \tag{21.24}$$

whereby

$$\frac{\bar{N}_{++}\bar{N}_{--}}{(\bar{N}_{+-})^2} = \frac{1}{4} e^{4\gamma} = \frac{1}{4} e^{4J/kT} \tag{21.25}$$

The last result differs significantly from the one that followed from the random mixing approximation. The difference lies in the extra factor $\exp(4J/kT)$ which, for $J > 0$, favors the formation of parallel-spin pairs $\uparrow\uparrow$ and $\downarrow\downarrow$, as opposed to antiparallel-spin pairs $\uparrow\downarrow$ and $\downarrow\uparrow$. In fact, one may regard the elementary process

$$\uparrow\uparrow + \downarrow\downarrow <=> 2\uparrow\downarrow \tag{21.26}$$

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-})_{\bar{L}=0} = \frac{1}{2}qN\frac{(e^{\gamma}, e^{\gamma}, 2e^{-\gamma})}{4\cosh\gamma} \tag{21.27}$$

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-})_{\bar{L}=0} = \frac{1}{2}q\frac{(1,1,2)}{4} \tag{21.28}$$

Now we evaluate the configurational energy U_0 and the specific heat C_0 of the lattice in the absence of the external field ($\alpha = 0$)

$$U_0 = -j(\frac{1}{2}qN - 2q\bar{N}_{+} + 4\bar{N}_{++})_{\alpha=0} \tag{21.29}$$

$$(\bar{N}_{+})_{\alpha=0} = \frac{1}{2}N(1+\bar{L}_0) = \frac{1}{2}N\frac{\exp(2\alpha') + \exp(-2\gamma)}{\cosh(2\alpha') + \exp(-2\gamma)} \tag{21.30}$$

$$U_0 = -\frac{1}{2}qJN\frac{\cosh(2\alpha') - \exp(-2\gamma)}{\cosh(2\alpha') + \exp(-2\gamma)} \tag{21.31}$$

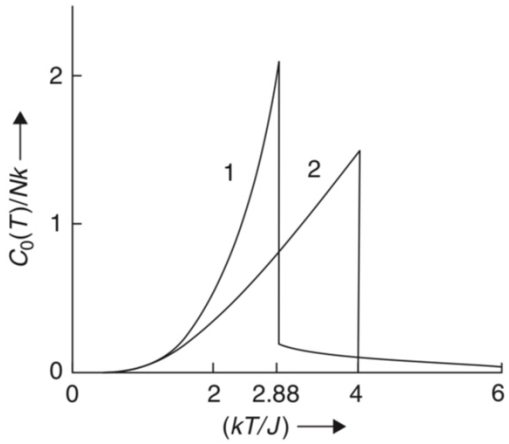


Figure 21.1: The field-free specific heat of an Ising lattice with coordination number 4. Curve 1 obtains in the Bethe approximation, curve 2 in the Bragg-Williams approximation.

$$U_0 = -\frac{1}{2}qJN\frac{1-\exp(-2\gamma)}{1+\exp(-2\gamma)} = -\frac{1}{2}qjN\tanh\gamma \tag{21.32}$$

$$C_0/Nk = \frac{1}{2}q\gamma^2\mathrm{sech}^2\gamma \quad (T > T_c) \tag{21.33}$$

$$\frac{1}{Nk}C_0(T_c+) = \frac{1}{2}q\gamma_c^2\text{sech}^2\gamma_c = \frac{1}{8}\frac{q^2(q-2)}{(q-1)^2}\left\{\ln\left(\frac{q}{q-2}\right)\right\}^2 \tag{21.34}$$

$$\begin{aligned} U_0 &= -\frac{1}{2}qJN\left[\frac{1}{q-1} + \frac{q(q-2)}{(q-1)^2}(-\gamma-\gamma_c) + \frac{q(q-2)}{(q-1)^2}\alpha'^2 + \cdots\right] \\ &= -\frac{1}{2}qJN\left[\frac{1}{q-1} + \frac{q(q-2)(3q-2)}{(q-1)^2}\frac{J}{kT_c}\left(1-\frac{T}{T_c}\right) + \cdots\right] \end{aligned} \tag{21.35}$$

$$\frac{1}{Nk}C_0(T_c-) = \frac{1}{8}\frac{q^2(q-2)(3q-2)}{(q-1)^2}\left\{\ln\left(\frac{q}{q-2}\right)\right\}^2 \tag{21.36}$$

In passing, we note that, according to Eq. 21.14, the transition temperature for a lattice with $q = 2$ is zero, which essentially means that a one-dimensional Ising chain does not undergo a phase transition. This result is in complete agreement with the one following from an exact treatment of the one-dimensional lattice; see Section 13.2. In fact, **for a lattice with $q = 2$, any results following from the Bethe approximation are completely identical with the corresponding exact results (see Problem 13.3); on the other hand, the Bragg-Williams approximation is least reliable when $q = 2$**.

That T_c for $q = 2$ is zero (rather than $2J/k$) is in line with the fact that, for all q, the first approximation yields a transition temperature closer to the correct value of T_c than does the zeroth approximation.

One important virtue of the Bethe approximation is that it brings out the role of the dimensionality of the lattice in bringing about a phase transition in the system. The fact that $T_c = 0$ for $q = 2$ and thereon it increases steadily with q leads one to infer that, while **a linear Ising chain does not undergo phase transition at any finite T, higher dimensionality does promote the phenomenon**. One may, in fact, argue that the absence of a phase transition in a one-dimensional chain is essentially due to the fact that, the interactions being severely short-ranged, communication between any two parts of the chain can be completely disrupted by a single defect in-between. The situation remains virtually unaltered even if the range of interactions is allowed to increase so long as it remains finite.